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Current Implementation and Barriers to Using Blood Flow Restriction Training: Insights From a Survey of Allied Health Practitioners

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Abstract

Scott, BR, Marston, KJ, Owens, J, Rolnick, N, and Patterson, SD. Current implementation and barriers to using blood flow restriction training: Insights from a survey of allied health practitioners. *J Strength Cond Res* XX(X): 000–000, 2023—This study investigated the use of blood flow restriction (BFR) exercise by practitioners working specifically with clinical or older populations, and the barriers preventing some practitioners from prescribing BFR. An online survey was disseminated globally to allied health practitioners, with data from 397 responders included in analyses. Responders who had prescribed BFR exercise (n = 308) completed questions about how they implement this technique. Those who had not prescribed BFR exercise (n = 89) provided information on barriers to using this technique, and a subset of these responders (n = 22) completed a follow-up survey to investigate how these barriers could be alleviated. Most practitioners prescribe BFR exercise for musculoskeletal rehabilitation clients (91.6%), with the BFR cuff pressure typically relative to arterial occlusion pressure (81.1%) and implemented with resistance (96.8%) or aerobic exercise (42.9%). Most practitioners screen for contraindications (68.2%), although minor side effects, including muscle soreness (65.8%), are common. The main barriers preventing some practitioners from using BFR are lack of equipment (60.2%), insufficient education (55.7%), and safety concerns (31.8%). Suggestions to alleviate these barriers included developing educational resources about the safe application and benefits of BFR exercise (n = 20) that are affordable (n = 3) and convenient (n = 4). These results indicate that BFR prescription for clinical and older cohorts mainly conforms with current guidelines, which is important considering the potentially increased risk for adverse events in these cohorts. However, barriers still prevent broader utility of BFR training, although some may be alleviated through well-developed educational offerings to train practitioners in using BFR exercise.

Key Words: KAATSU, occlusion, rehabilitation, sports medicine, strength

Introduction

Exercise with blood flow restriction (BFR) is a technique that involves applying inflatable cuffs proximally on the limbs during exercise to partially decrease arterial inflow and occlude venous return (15). Altering hemodynamics increases the physiological stress of exercise, eliciting potent adaptations while only requiring low mechanical loads (28). For example, adding BFR during low-load resistance training (e.g., 20–50% 1 repetition maximum [1RM]) increases muscle size and strength similarly to unrestricted heavy-load training (e.g., >60% 1RM) (10,14), and low-intensity aerobic training (e.g., walking at 3–5 km·h^{-1}) with BFR enhances cardiovascular fitness and vascular adaptations that normally require vigorous training (1). Therefore, training with BFR may benefit individuals who are unable to tolerate the mechanical demands of strenuous exercise, such as those suffering from disease (5), recovering from injury (4), or older adults (2). As such, BFR training has increased in popularity over the past 20 years. However, the prescription of BFR exercise has varied across studies, which may cause confusion for practitioners. Although recent BFR training guidelines are available (22), there are limited recent data regarding the real-world prescription of BFR exercise, particularly for clinical populations and older people.

Survey results of facilities offering KAATSU training (specific BFR training methodology developed in Japan) have indicated that this technique is prescribed in a range of treatment environments for patients of all ages and to target improvements in several categories (including orthopedic issues, sporting performance, obesity, and physical health) (20,34). Low incidence of adverse side effects was also presented, with subcutaneous hemorrhage (13.1%) and numbness (1.3%) being the most common (20). However, these surveys were conducted several years ago by the inventor and founder of KAATSU equipment and focused on KAATSU training facilities located in Japan. Taking a global approach, Patterson and Brandner (21) surveyed 250 practitioners of whom 115 had implemented BFR exercise. This study provided comprehensive information regarding how BFR was prescribed with resistance and aerobic exercise and for passive applications. The incidence of some minor side effects was higher than those reported by Nakajima et al. (20), with delayed-onset muscle soreness, numbness, and fainting/dizziness reported by ~39, ~18, and ~15% of responders, respectively. However, the subjects in this survey worked primarily in strength and conditioning and research roles (combined 64% of responders),

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meaning that limited inferences could be made about the use of BFR for more clinical populations.

Considering that BFR training likely has the greatest utility for individuals who cannot tolerate the mechanical demands of vigorous activity (24), it is perhaps more meaningful to understand how the technique is implemented by allied health professionals working with such populations (i.e., older people, those with clinical conditions or recovering from injury). Safety concerns regarding BFR typically focus on these populations, who may be at an increased risk of adverse side effects (29). Information regarding the prevalence of adverse reactions to BFR exercise for those working with these "at-risk" cohorts is needed. Furthermore, as safety concerns are recognized as a barrier to prescribing BFR exercise (27), it is also important to understand factors that prevent allied health practitioners from using this intervention. Preliminary research indicates that the barriers to using BFR include equipment cost, lack of training, lack of resources, and concerns with clinical efficacy and safety (6,18). However, these preliminary surveys were not focused on using BFR specifically for clinical or older cohorts, have not reported practitioner perceptions on how these barriers could be alleviated, and only included practitioners in the United States. If common barriers to BFR use can be alleviated, this technique may become more widely used for clinical and older cohorts.

Therefore, the primary purpose of this study was to investigate the current global utilization of BFR exercise by allied health practitioners working specifically with clinical or older populations. This study also aimed to report on the incidence of side effects to BFR exercise in these populations, and the preintervention screening processes used by practitioners. Finally, the research aimed to identify barriers preventing some practitioners from prescribing BFR exercise and approaches that may help overcome these barriers.

Methods

Experimental Approach to the Problem

A self-administered online questionnaire was developed to survey allied health practitioners on their usage of BFR exercise. Inclusion criteria required subjects to be aged older than 18 years, currently involved in exercise prescription or delivery for clinical or older populations, be aware of BFR exercise, and to have sufficient competency with English language. This primary survey was structured so that responders who indicated that they have prescribed BFR exercise for their clients were asked questions about how they implement this technique, whereas those who reported not prescribing BFR exercise were asked to identify the barriers to using this exercise. A subset of those who responded that they had not used BFR also agreed to receive a follow-up survey to investigate how these barriers could be alleviated. This advertisement of the primary survey occurred from April 2021 until February 2022 through social media, email, and online forums (e.g., newsletters of professional or accrediting organizations relating to fields such as exercise physiology, physiotherapy, rehabilitation, etc). The follow-up survey was sent directly to the subset of responders who agreed to complete this through email between June and August 2022.

Subjects

The survey was advertised to allied health practitioners involved in exercise for clinical, injured, and older populations (including exercise physiologists, physiotherapists, rehabilitation specialists, researchers, strength and conditioning coaches, personal trainers, sport scientists, and medical doctors; Table 1 for practitioner demographics). Before beginning the survey, all subjects completed an online informed consent form. This research and its methods were approved by the Murdoch University Institutional Human Research Ethics Committee.

A total of 520 responders provided informed consent to complete the survey. Of these, 61 did not complete any further questions, and 47 had not heard of BFR. Nine reported using BFR but provided no information on how they do so, and 6 reported not using BFR but answered no questions on barriers to the technique. Following these exclusions, 397 responders were aware of BFR being used for training or rehabilitation and completed sufficient information for their data to be included in the analysis for the primary survey (have prescribed BFR: provided demographic details and at least information on the aims and equipment used for BFR exercise; have not prescribed BFR: demographic details and at least information on the barriers to BFR or

Table 1

Demographic details for practitioners who responded to the survey.*

	Have used BFR	Have not used BFR
Gender	*n* = 308	*n* = 89
Male	226 (73.4%)	49 (55.1%)
Female	82 (26.6%)	39 (43.8%)
Other/nonbinary	—	1 (1.1%)
Age distribution	*n* = 308	*n* = 89
18–29	81 (26.3%)	33 (37.1%)
30–39	141 (45.8%)	31 (34.8%)
40–49	60 (19.5%)	16 (18.0%)
50–59	19 (6.2%)	6 (6.7%)
≥60	7 (2.3%)	3 (3.4%)
Place of residence	*n* = 307	*n* = 89
North America	214 (69.7%)	51 (57.3%)
Europe	45 (14.7%)	18 (20.2%)
Australia and New Zealand	22 (7.2%)	9 (10.1%)
United Kingdom	15 (4.9%)	7 (7.9%)
Asia, Africa, and Middle East	11 (3.6%)	4 (4.5%)
Current profession	*n* = 312†	*n* = 89
Physiotherapist or physical therapist	239 (76.6%)	66 (74.2%)
Athletic trainer	18 (5.8%)	2 (2.2%)
Exercise physiologist	17 (5.4%)	7 (7.9%)
Researcher	7 (2.2%)	1 (1.1%)
Sport scientist	7 (2.2%)	—
Chiropractor	5 (1.6%)	—
Personal trainer	4 (1.3%)	5 (5.6%)
Medical doctor	4 (1.3%)	1 (1.1%)
Rehabilitation specialist	3 (1.0%)	—
Strength and conditioning coach	2 (0.6%)	1 (1.1%)
Other	6 (1.9%)	6 (6.7%)
Years worked in profession	*n* = 307	*n* = 88
<10 y	183 (59.6%)	55 (62.5%)
10–20 y	75 (24.4%)	18 (20.5%)
>20 y	49 (16.0%)	15 (17.0%)
Primary work setting	*n* = 308†	*n* = 89
Private clinic or facility	171 (55.5%)	53 (59.6%)
Sporting environment	66 (21.4%)	9 (10.1%)
Hospital or outpatient	34 (11.0%)	13 (14.6%)
University or research facility	30 (9.7%)	7 (7.9%)
Commercial gym	1 (0.3%)	3 (3.4%)
Military	1 (0.3%)	1 (1.1%)
Other	5 (1.6%)	3 (3.4%)

*Data represented as the number (percentage) of respondents who reported prescribing blood flow restriction (BFR) exercise for their clients and those who have not prescribed BFR. The option for "other" was provided for situations where prewritten options were not appropriate for a responder. When selecting "other," responders were asked to provide specific details by means of a free-text field, and these were included in the table if related to existing categories or as new categories if reported by more than 1 responder.

†Some practitioners from indicated 2 responses through free text.

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willingness to undertake education on BFR). Of these, 308 (77.6%) had previously prescribed BFR exercise for their clients. An additional 89 subjects (22.4%) were aware of it but had not used BFR. Of the 72 responders who agreed to receive a follow-up survey on the barriers to BFR exercise, 27 completed sufficient information to be included. Five (18.5%) of these reported that they had begun to use BFR since completing the primary questionnaire after overcoming barriers, including lack of equipment, safety concerns, and no training in BFR. The remaining 22 responders answered questions on alleviating barriers to BFR exercise.

Procedures

Survey data were collected using online questionnaire software (Qualtrics Provo, UT; see Supplementary File, Supplemental Digital Content, http://links.lww.com/JSCR/A437). The primary survey contained 8 sections, comprising 38 open and closed questions to understand how BFR exercise is being implemented by allied health practitioners and the barriers preventing its implementation; (1) subject demographics, (2) types of clients and goals of BFR training, (3) how BFR is implemented, (4) prescription of resistance exercise or (5) aerobic exercise, (6) the safety and contraindications to BFR training, (7) the main barriers to prescribing BFR training, and (8) willingness to engage in educational opportunities on BFR exercise. Subjects who reported that they have prescribed BFR exercise for clients completed sections 1–6, whereas those who reported not using BFR exercise completed sections 1 and 7–8. Subjects who completed the follow-up on barriers to prescribing BFR exercise responded to 25 open and closed questions to identify (1) whether they were now using BFR, and if not, (2) the main concerns regarding the 3 most prevalent barriers identified from the primary survey, and (3) what strategies could help overcome these barriers.

In accordance with Ebben and Blackard (8), the questionnaire was developed by the authors and pilot tested with an informal advisory group of 3 allied health practitioners (exercise physiologists) with varying knowledge of BFR to ensure clarity of the questions. After reviewing the survey responses provided by the advisory group members in conjunction with the feedback they gave on survey clarity and structure, some questions were amended before the survey was considered ready for distribution.

Statistical Analyses

Content analysis was done to code free-text responses from open questions into concepts related to each question. Descriptive statistics (counts and frequencies) were calculated from this analysis and for responses to closed questions. Chi-squared (χ^2) tests of contingencies were used to determine whether the use of BFR was influenced by practitioner or client demographics or whether client demographics impacted the incidence of side effects to BFR. Data management and analysis were performed using Microsoft Excel (v16.61 Microsoft Corp., Washington, WA) and SPSS (v27, Chicago, IL).

Table 2

Description of clients with whom blood flow restriction (BFR) exercise is prescribed.*

	Absolute responders	Relative responders (%)
Clients with which BFR is used (n = 308)†		
Musculoskeletal/rehabilitation	282	91.6
Healthy younger people	233	75.6
Healthy older people	157	51.0
Sarcopenic clients	64	20.8
Chronic disease	62	20.1
Client age range (n = 308)†		
<20 y	167	54.2
21–30 y	259	84.1
31–40 y	214	69.5
41–50 y	177	57.5
51–60 y	137	44.5
61–70 y	80	26.0
71–80 y	33	10.7
80+ y	10	3.2
Aims of BFR training (n = 308)†		
Muscle strengthening	266	86.4
Injury rehabilitation	257	83.4
Muscle hypertrophy	241	78.2
Avoiding loss of muscle mass	204	66.2
Bone strengthening	63	20.5
Improving cardiovascular fitness	59	19.2
Improving general health	50	16.2
Pain reduction	5	1.6
Recovery	2	0.6
Other	4	1.3

*The option for "other" was provided for situations where prewritten options were not appropriate for a responder. If "other" was selected, responders were asked to provide specific details by means of a free-text field, and these were included in the table if related to existing categories or as new categories if reported by more than 1 responder.
†Subjects could select all which applied.

Table 3

Parameters for the prescription of blood flow restriction (BFR) during resistance and aerobic exercise.*†

	Absolute responses	Relative responses (%)
Determination of cuff pressure (n = 307)		
% measured LOP or AOP	249	81.1
Perceived tightness or pressure	14	4.6
Arbitrary pressure	13	4.2
% resting blood pressure	7	2.3
% estimated LOP or AOP§	6	2.0
% limb circumference	3	1.0
Other	15	4.9
Cuff system used (n = 309)‡		
Automatic inflatable tourniquet	186	60.2
Handheld inflatable pump	108	35.0
Elastic knee wraps	7	2.3
Kaatsu training device	2	0.6
Other	6	1.9
Continuous or intermittent cuff application (resistance exercise; n = 272)		
Continuous (including rest)	208	76.5
Intermittent (excluding rest)	27	9.9
Depends on situation	38	14.0
BFR used unsupervised (n = 266)		
Yes	53	19.9
No	213	80.1
Frequency of training done unsupervised (n = 53)		
0 days per week	2	3.8
1 day per week	12	22.6
2 days per week	23	43.4
3 days per week	10	18.9
4 days per week	3	5.7
5 days per week	3	5.7

*LOP = limb occlusion pressure; AOP = arterial occlusion pressure.
†The option for "other" was provided for situations where prewritten options were not appropriate for a responder. If "other" was selected, responders were asked to provide specific details by means of a free-text field, and these were included in the table if related to existing categories or as new categories if reported by more than 1 responder.
‡One responder indicated 2 different systems in free text.
§Estimated through established equations.

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Results

Practitioner and Client Demographics

Demographic details for the 397 subjects included in the analysis are presented in Table 1, with absolute and relative response rates reported for the 2 distinct groups (i.e., those who have and who have not prescribed BFR exercise). Information regarding the types of clients with whom BFR is typically prescribed is shown in Table 2.

Regarding the associations between BFR use and practitioner or client demographics, a greater proportion of practitioners who have prescribed BFR exercise reported working with musculoskeletal rehabilitation clients (84.4%) than not working with these rehabilitation clients (15.6%; $\chi^2 = 6.03$, $p = 0.014$). A lower proportion of practitioners who have prescribed BFR exercise reported working with chronic disease clients (26.3%) than not (73.7%; $\chi^2 = 3.92$, $p = 0.048$). No significant associations were observed between other practitioner demographics (place of residence, profession, or primary work setting) or client demographics (healthy young, healthy older, or sarcopenic clients) and the use of BFR.

Applications of Blood Flow Restriction

Table 3 presents an overview of how practitioners have prescribed BFR during exercise. Most responders base BFR pressure on an assessment of arterial occlusion pressure (AOP). Seventeen responders also provided free-text information on the relative pressure used, which was primarily in the range of 60–80% AOP lower body ($n = 16$) and 40–60% AOP for the upper body ($n = 15$). Four responders indicated that they use perceptual responses to apply BFR with pressure ratings of 6–8 and 5–7 out of 10 for the lower and upper body, respectively.

Table 4

Exercise prescription parameters categorized from free-text responses for resistance exercise ($n = 298$) and aerobic exercise ($n = 132$).*

Resistance exercise ($n = 298$)		Aerobic exercise ($n = 132$)	
Category	***n* (%)**	**Category**	***n* (%)**
Exercises used (n = 272)		**Exercises used (n = 127)**	
Multijoint upper body	54 (19.9)	Cycling	111 (87.4)
Singlejoint upper body	124 (45.6)	Treadmill or walking	72 (56.7)
Singlejoint lower body	205 (75.4)	Upper-body ergometer	11 (8.7)
Multijoint lower body	221 (81.3)	Stepper or elliptical	8 (6.3)
		Rowing ergo	4 (3.1)
		Running	3 (2.4)
Number of sets per exercise (n = 272)		**Intensity prescription or monitoring technique (n = 121)**	
1	3 (1.1)	Perception	34 (28.1)
2	15 (5.5)	Heart rate	33 (27.3)
3	69 (25.4)	Speed or cadence	22 (18.2)
4	241 (88.6)	Oxygen consumption	8 (6.6)
5	23 (8.5)	Power output	5 (4.1)
>5	4 (1.5)		
Number of repetitions (n = 272)		**Exercise intensity implemented (n = 121)†**	
<10	14 (5.1)	Low	37 (30.6)
10–15	61 (22.4)	Moderate	37 (30.6)
16–20	41 (15.1)	High	12 (9.9)
>20	37 (13.6)		
30-15-15-15 scheme	183 (67.3)		
Rest periods (n = 272)		**Interval or continuous exercise (n = 127)**	
<30 s	5 (1.8)	Interval	52 (40.9)
30–45 s	204 (75.0)	Continuous	109 (85.8)
46–60 s	94 (34.6)		
>60 s	32 (11.8)		
Load lifted (n = 270)		**Duration of blood flow restriction exercise (n = 125)**	
≤19% maximum	6 (2.2)	<10 min	57 (45.6)
20–29% maximum	56 (20.7)	10–20 min	96 (76.8)
30–39% maximum	67 (24.8)	21–30 min	13 (10.4)
40–49% maximum	24 (8.9)	>30 min	1 (0.8)
50–59% maximum	12 (4.4)		
≥60% maximum	7 (2.6)		
Training to failure (n = 270)		**Training to failure (n = 127)**	
Yes	102 (37.8)	Yes	12 (9.4)
No	75 (27.8)	No	98 (77.2)
Sometimes	90 (33.3)	Sometimes	17 (13.4)
Training frequency (days per week; n = 272)		**Training frequency (days per week; n = 125)**	
1 d	57 (21.0)	1 d	36 (28.8)
2–3 d	260 (95.6)	2–3 d	114 (91.2)
4–6 d	21 (7.7)	4–6 d	13 (10.4)
Daily	2 (0.7)	Daily	1 (0.8)

*Data represented as the absolute and relative responses based on the sample who responded to questions related to these categories. Responses from some subjects either did not fit into the categories in this table (e.g., reported "depends on client") or included multiple categories in 1 response (e.g., use both singlejoint and multijoint lower-body exercise).

†Categorized from free-text responses based on relative intensity definitions from the American College of Sports Medicine (9).

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Prescription of Exercise with Blood Flow Restriction

Table 4 describes the exercise prescription parameters of BFR training reported by subjects. Prescribing resistance training with BFR was more common than aerobic training modalities ($n = 298$ and 132, respectively). Numerous different resistance exercises are used by practitioners, with the most common being the squat (used by 61.0% of responders), leg extension (57.5%), leg press (43.0%), bicep curl (36.4%), leg curl (26.1%), calf raise (25.7%), and deadlift (19.1%).

Contraindications and Side Effects

Most subjects reported using a specific screening tool to determine whether BFR is suitable for a client (182 of 267 responses, 68.2%). Subjects were asked to respond to a series of potential contraindications that are typically included in pre-BFR exercise screening tools (12) and identify whether they would use BFR in these circumstances (Figure 1A). Furthermore, subjects were asked to identify any other conditions preventing them from applying BFR or prompt them to seek medical clearance (Figure 1B). Subjects also indicated whether their clients had experienced some commonly reported side effects from BFR (Figure 2). Additional side effects reported by means of free text by multiple subjects included itching ($n = 4$), petechiae ($n = 3$), paresthesia ($n = 3$), and tingling ($n = 2$). No responders reported their clients experiencing myocardial infarction or rhabdomyolysis.

Clients experiencing dizziness were reported by more practitioners who prescribe BFR for people aged >61 years (33.3%) compared with those who do not (17.2%; $\chi^2 = 9.25$, $p = 0.002$), more practitioners working with chronic disease clients (33.9%) than those who do not (18.3%; $\chi^2 = 7.14$, $p = 0.008$), more practitioners working with musculoskeletal rehabilitation clients (23.0%) than those who do not (3.8%; $\chi^2 = 5.21$, $p = 0.022$), and more practitioners working with

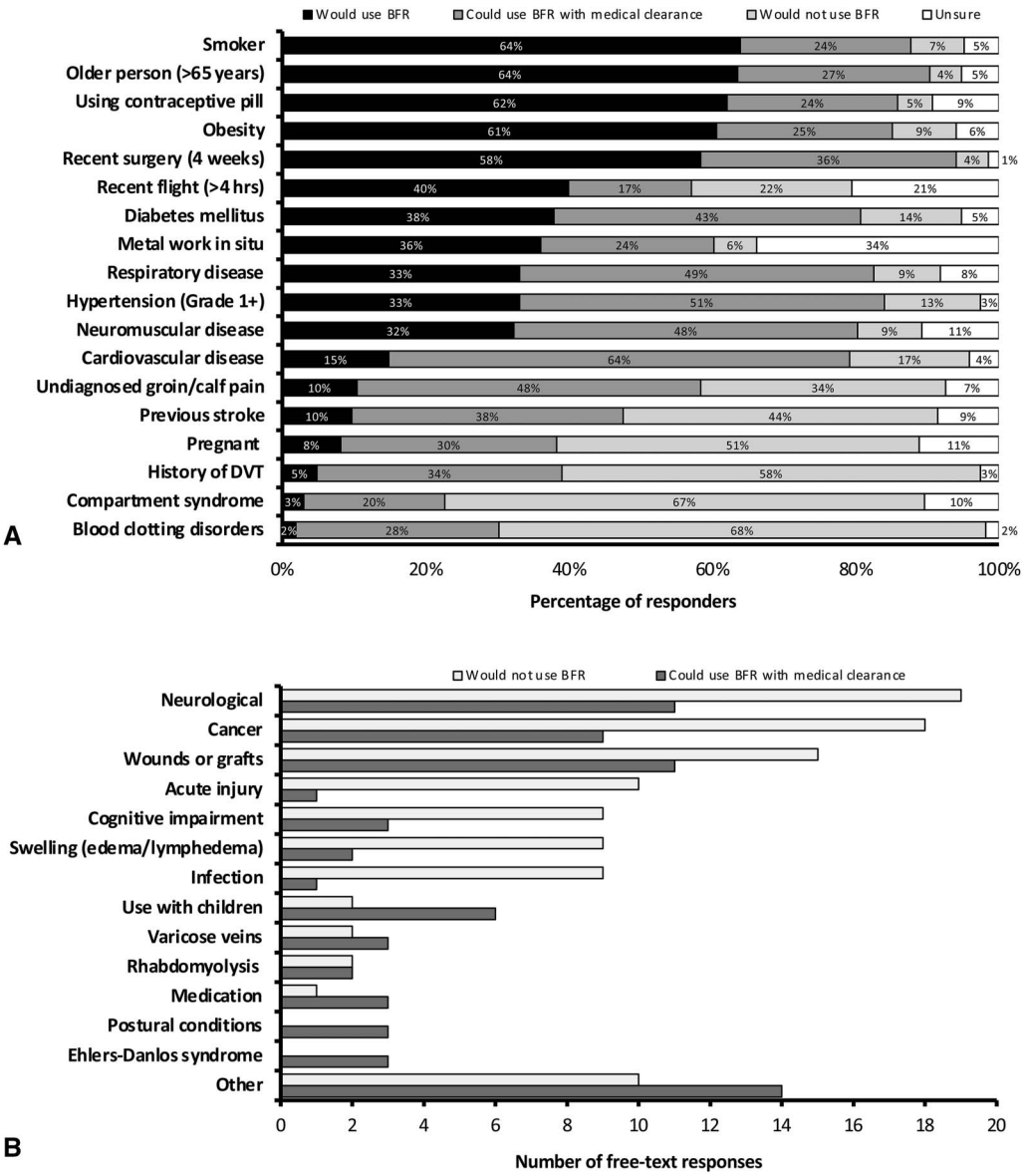

Figure 1. Responses to the question "would you use blood flow restriction (BFR) or seek further medical approval for clients with these circumstances?" (A; $n = 269$), and additional considerations and contraindications for using BFR highlighted by practitioners in free-text responses (B). Additional considerations that were not included by more than 1 responder are grouped as "other." DVT = deep vein thrombosis.

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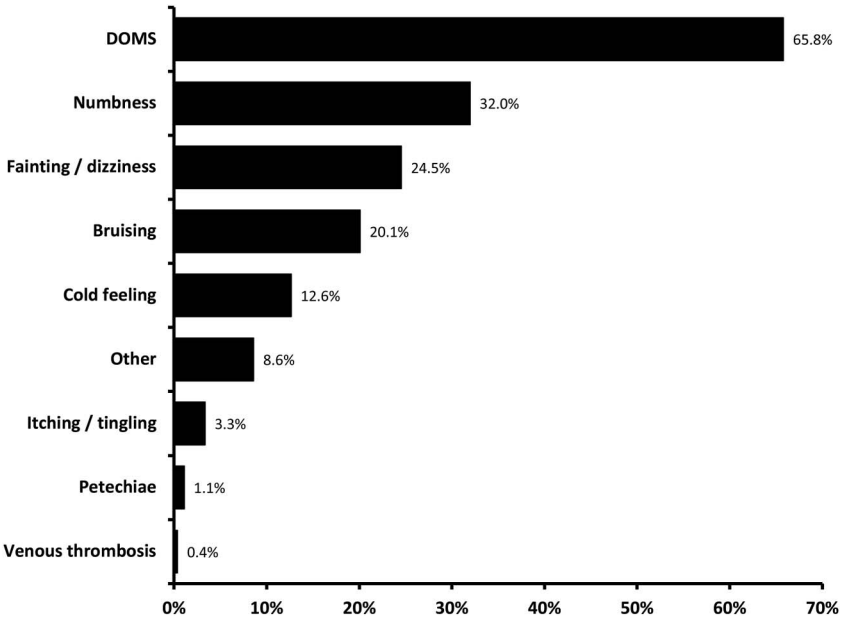

Figure 2. Percentage of responders who reported side effects to blood flow restriction (BFR) training in their clients (n = 269). The option for "other" was provided for situations where prewritten options were not appropriate for a responder. If "other" is selected, responders were asked to provide specific details by means of a free-text field, and these were included in the figure if reported by more than 1 responder. DOMS = delayed-onset muscle soreness.

sarcopenic clients (31.3%) than those who do not (18.9%; $\chi^2 = 4.63$, $p = 0.031$). Clients experiencing numbness from BFR were reported by more physiotherapists or physical therapists (31.1%) compared with other professions (17.8%; $\chi^2 = 4.86$, $p = 0.027$), and experiencing fainting or dizziness from BFR was more common in North America (24.8%) compared with other locations (14.0%; $\chi^2 = 4.47$, $p = 0.034$). No significant associations were observed between client demographics or geographical location and the incidence of other minor side effects.

Barriers to Prescribing Blood Flow Restriction Exercise

Several barriers preventing the prescription of BFR exercise were identified by responders who reported not using BFR for their clients (Figure 3). The 3 most common barriers (lack of equipment, insufficient training, and safety concerns) were further investigated in the follow-up survey.

Of the 22 responders to the follow-up survey who still had not prescribed BFR, 50.0% had explored different types of cuff systems to find affordable alternatives, and 72.7% indicated that they would

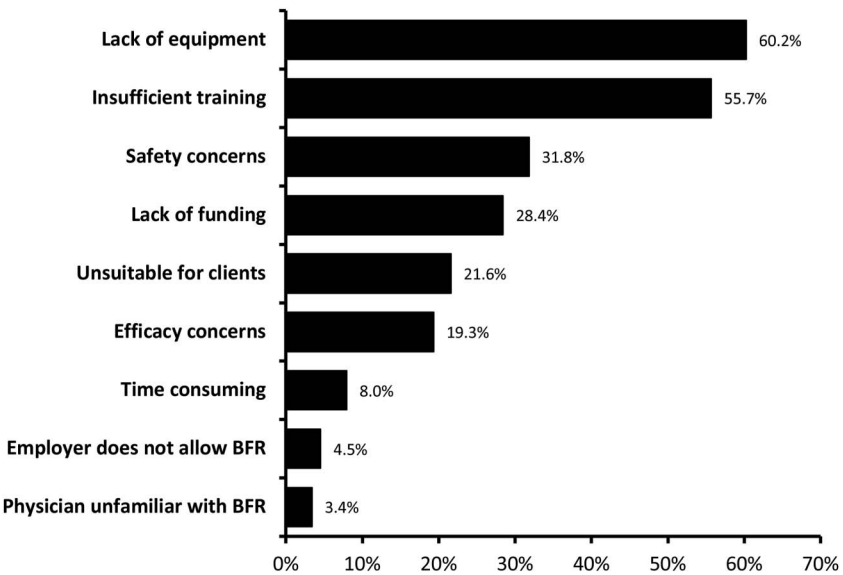

Figure 3. Percentage of responders who reported barriers that prevented them from prescribing blood flow restriction (BFR) training for their clients (n = 88). Responders could select more than 1 response or provide free-text information.

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consider prescribing BFR exercise if they had access to the equipment. Most responders indicated that they would be interested in BFR educational opportunities (Figure 4A; primary survey), through online and face-to-face modalities. The follow-up survey also asked responders to indicate the topics which would be most important to them in an educational offering (Figure 4B). The main safety concerns raised in the follow-up survey included risk of blood clots or thrombosis (n = 14, 63.6% or responders) or cardiac events (n = 11, 50.0%), followed by acute pain, stroke, and nerve or vascular injury (all n = 10, 45.5%).

Responders suggested several approaches to overcoming the main barriers to BFR exercise. To overcome barriers associated with equipment access, gaining the support of employers (n = 9) and access to affordable or rental equipment (n = 5) were the most common requirements. To alleviate the barriers of insufficient education and safety concerns, several responders suggested educational resources about the safe application and benefits of BFR exercise (n = 20), which are affordable (n = 3) and convenient (e.g., online; n = 4).

Discussion

This study recruited allied health practitioners working with clinical or older populations to investigate the current utilization of BFR exercise, incidence of side effects, screening processes being used, and the barriers preventing some practitioners from prescribing BFR. The key findings are as follows: (1) most practitioners prescribe BFR exercise for musculoskeletal rehabilitation clients to increase muscle strength and size, (2) cuff pressure is typically based on AOP assessment and implemented with low-load resistance exercise or low-to-moderate–intensity aerobic exercise, (3) most practitioners screen clients for contraindications, although minor side effects including muscle soreness are common, and (4) the most common barriers preventing some practitioners from using BFR are lack of equipment, insufficient education, and safety concerns. These results indicate that BFR prescription for clinical and older clients mainly conforms with current guidelines (22), although barriers prevent broader use of BFR training.

The responders to this survey were primarily physiotherapists or physical therapists (76.6%), reflecting our targeted recruitment of subjects working with clinical, rehabilitation, or older cohorts. This demographic differs from that of Patterson and Brandner (21), where only 23.6% of responders worked as physiotherapists, rehabilitation specialists, or medical doctors. Considering our practitioner demographic, it is not surprising that 91.6% use BFR primarily for musculoskeletal rehabilitation focused on increasing muscle strength

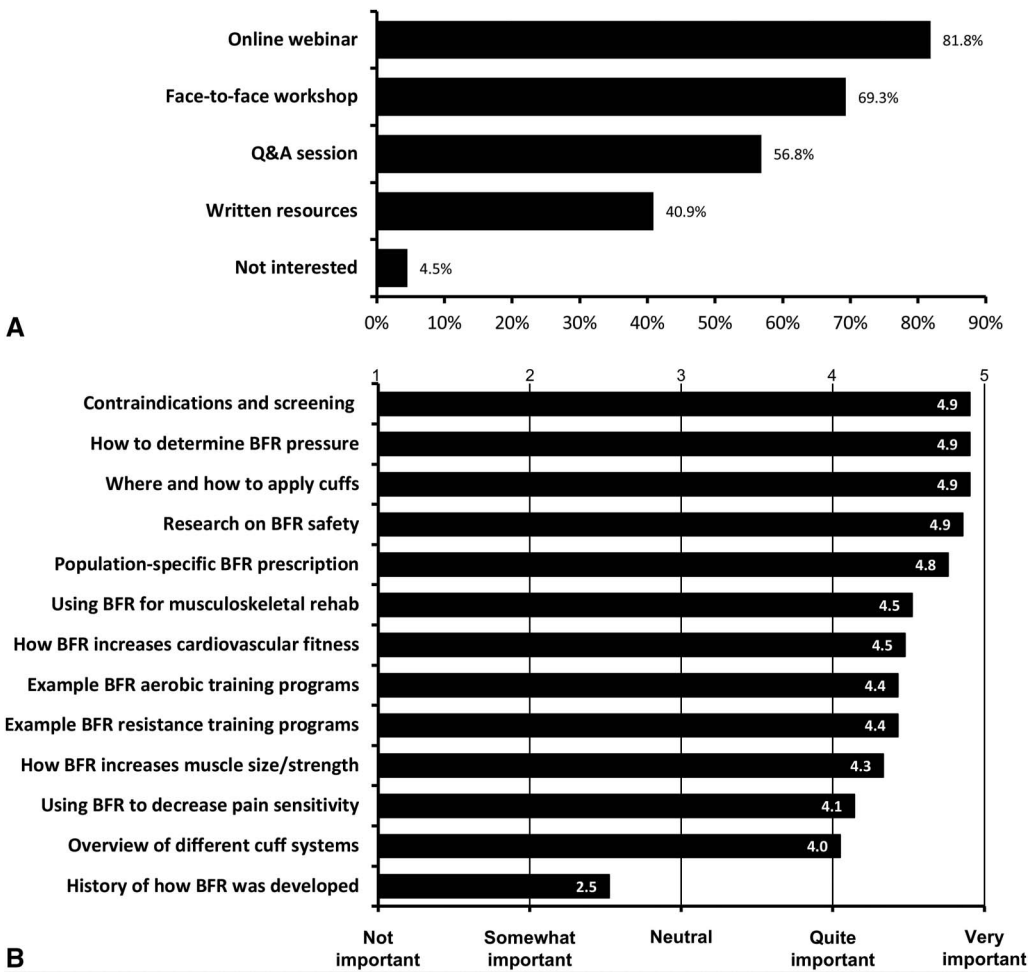

Figure 4. Preferences for types of educational activities from practitioners who reported barriers to prescribing blood flow restriction (BFR) training (n = 88; A) and for key topics to be covered in educational activities (n = 21; B). Q&A = question and answer.

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and size (86.4 and 78.2% of responders, respectively), for injury rehabilitation (83.4%), and to avoid losing muscle mass (66.2%). These results were supported by chi-squared tests, with a greater proportion of responders who prescribe BFR exercise working with musculoskeletal rehabilitation clients than not working with these clients.

There is a growing body of literature to support the utility of BFR exercise for specific patients, including those with multiple sclerosis (13), rheumatoid arthritis (25), sporadic inclusion body myositis (11), chronic kidney disease (26), and following anterior cruciate ligament surgery (30), along with promising case reports for patients with Parkinson's disease (7) and sarcopenia (17). Interview and focus group data indicate that patients with chronic obstructive pulmonary disease and their health care professionals have positive perceptions of using BFR exercise for health benefits; however, further evidence of the suitability and appropriate prescription of BFR for specific conditions is still required (23).

Early BFR research prescribed cuff pressure at arbitrary values across all subjects (19). Survey data collected in 2015 indicated that 43.4% of practitioners prescribed BFR based on preassigned pressures from the literature, whereas only 11.5% of responders used a percentage of AOP (21). Generally, researchers agree that applying the BFR relative to AOP is most appropriate because it accounts for interindividual differences in the hemodynamic responses during BFR exercise (22). Our data indicate that BFR pressure is now primarily prescribed relative to measured AOP (81.1% of responders), with only 4.2% using arbitrary pressures. This may be partly the result of most responders (60.2%) also using automatic inflatable tourniquet systems that often incorporate assessment of the AOP into their BFR programs. However, 35.0% of responders used handheld inflatable pump systems, which require a Doppler ultrasound device for robust assessment of the AOP. Of the free-text responses regarding the actual pressures used, practitioners primarily implement 50–80% AOP for the lower body and 40–60% AOP for the upper body, which aligns with current recommendations (22).

More responders prescribe BFR during resistance exercise (n = 298) than aerobic exercise (n = 132). In agreement with Patterson and Brandner (21), BFR was more commonly used for lower-body than for upper-body resistance exercises, with both singlejoint and multijoint movements being prescribed. Although only the limb muscles are exposed to a restrictive stimulus during BFR, bench press exercise with BFR acutely increases pectoralis major activation (33) and promotes larger improvements in pectoralis major thickness over 2 weeks of twice daily training compared with unrestricted exercise (32), indicating some merit to multijoint exercise with BFR. Prescription of BFR resistance training in this survey aligns with recommendations: loads of 20–39% maximum strength are most commonly used, predominantly for 4 sets following the 30-15-15-15 repetition scheme with 30–45 seconds of rest between the sets, cuffs remaining inflated between the sets, and training frequencies of 2–3 days per week. Interestingly, 71.1% of responders reported that their clients perform sets to failure some of the time. Free-text responses indicated that this was sometimes intentional to provide a maximal stimulus in the final set(s) of BFR exercise, but it also occurred unexpectedly in some clients (i.e., set was not intended to cause failure, but client could not complete prescribed repetitions).

The most common modalities of aerobic BFR training were cycling and treadmill or walking. Training was prescribed through a range of techniques (e.g., perception, heart rate, set speed, or cadence) and corresponded to low (30.6% of responders) and moderate (30.6%) intensity, based on relative intensity definitions from the American College of Sports Medicine (9). Aerobic BFR training was typically prescribed as continuous rather than interval exercise, for 10–20 minutes 2–3 days per week. More responders indicated that their clients do not perform aerobic BFR training to failure (77.2%) compared with resistance exercise (27.8%). Responders indicated that BFR exercise is typically not performed unsupervised by their clients, although 19.9% have prescribed unsupervised BFR training.

Most practitioners (68.2%) screen their clients to determine suitability for BFR. The most caution was taken with individuals who have a blood clotting disorder, compartment syndrome, history of deep vein thrombosis, are pregnant, or have had a previous stroke. Chi-squared tests also indicated that of the practitioners who use BFR exercise, there are more who do not work with chronic disease clients than do. These observations align with the screening tool published by Kacin et al. (12) where family or personal history of clotting disorders, deep vein thrombosis, and stroke are all classified as absolute risk factors (i.e., do not prescribe BFR exercise). Interestingly, grade 1 (or higher) hypertension is also classified as an absolute risk factor (12), yet 84% of responders indicated that they would prescribe BFR for these clients or could with medical clearance. It may be that practitioners have gained confidence from studies that have implemented BFR for hypertensive patients without reported adverse reactions (Wong et al. (31) for systematic review and meta-analysis). However, because BFR will activate the exercise pressor reflex and potentially exaggerate increases in sympathetic nerve activity hypertensive patients (29), we recommend caution in prescribing BFR for hypertensive clients.

Previous research has suggested that muscle damage, which precedes delayed-onset muscle soreness, is uncommon with BFR resistance training (16). However, 65.8% of our responders reported that their clients had experienced delayed-onset muscle soreness, which is higher than the 39% of responders from Patterson and Brandner (21). These dissimilar findings are likely because of our focus on BFR use by clinical and older cohorts; these clients are more likely to be unaccustomed to exercise, meaning that even light loads with BFR could present a challenging stimulus that causes muscle damage or soreness. As low doses of resistive activities are protective against muscle soreness from subsequent exercise because of the repeated bout effect (3), we recommend practitioners to gradually introduce BFR exercise with clinical and older clients to limit acute muscle soreness.

The most common barriers reported were a lack of BFR equipment, insufficient education on BFR, and safety concerns, which confers with previous research (6,18). Practitioners cannot use BFR if they have no access to equipment (60.2% of responders), often related to a lack of funding (28.4%). These barriers are challenging to overcome, with ways to gain the support from employers to use BFR (n = 9), and access to affordable or rental equipment (n = 5) being the most common suggestions from practitioners. Many practitioners also reported that insufficient training or education on BFR (55.7%) and safety concerns with the technique (31.8%) limit its use. Suggestions for addressing these barriers from free-text answers included educational resources about the safe application and benefits of BFR exercise (n = 20), which are affordable (n = 3) and convenient (e.g., online; n = 4).

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These suggestions align to the most preferred modality for training opportunities (online webinar; 81.8%). Many responders would also be willing to attend face-to-face workshops (69.3%), which would assist practitioners to gain experience with assessing the AOP and using BFR cuffs. It is worth noting that contraindications, screening, and the safety of BFR were all amongst the topics rated as the most important to include in any educational offerings (Figure 4B), meaning that well-designed training on BFR could alleviate 2 of the 3 most prominent barriers for this technique.

This is the largest survey published on the use of BFR exercise (n = 397), and the first to focus on practitioners working with clinical and older clients. Importantly, this target population is more likely to require low-load exercise alternatives, such as BFR, than those examined in previous surveys (20,21). This research is also strengthened by collecting practitioner perspectives on barriers, which prevent them from prescribing BFR exercise, as well as potential approaches to alleviate these barriers. Nevertheless, some limitations should also be acknowledged. Several questions provided responders with selections (e.g., a list of typical aims of BFR training) followed by an "other" option for open responses, which could have biased prewritten options. Conversely, some questions deliberately included only free-text response boxes (e.g., BFR pressure used, BFR prescription parameters), meaning that some questions had reduced response rates and responses from some practitioners were difficult to categorize. As most responses were collected from physiotherapists or physical therapists and practitioners in North America (with low or no responses from other countries), our findings may not be generalizable outside of this demographic, although our analyses observed limited associations between practitioner type or geographic location and the use or side effects of BFR.

In conclusion, of the practitioners surveyed, 91.6% prescribe BFR exercise for musculoskeletal rehabilitation clients for muscle strengthening (86.4%), injury rehabilitation (83.4%), and hypertrophy (78.2%). The BFR pressure was mostly prescribed relative to AOP (81.1%). Resistance training was mainly used with low-load exercises and the 30-15-15-15 repetition scheme, and aerobic training was implemented with several modes of low-to-moderate–intensity cardiovascular exercise. Practitioners screen for contraindications, including clotting disorders, compartment syndrome, or history of deep vein thrombosis, and two-thirds of responders indicated that their patients experience muscle soreness following BFR exercise. The common barriers preventing some practitioners from prescribing BFR exercise include limited access to equipment (60.2%), insufficient training on BFR (55.7%), and safety concerns (31.8%). Approximately 95% of these responders would be willing to undertake education on BFR to address some of these limitations.

Practical Applications

Allied health practitioners prescribing BFR exercise for clinical and older clients often screen for contraindications before using this technique, aligning with current evidence-based guidelines (22). Nevertheless, minor side effects to BFR exercise were still reported with greater frequency than in previous surveys, likely because of the specific client demographic targeted in this research. To reduce the likelihood of side effects, practitioners should gradually implement BFR exercise into a client's training program by beginning with lower cuff pressures and exercise volumes and increasing to the desired percentage of AOP and external demands over several sessions.

Although barriers prevent some practitioners from prescribing BFR exercise, some of these could be alleviated by well-developed BFR educational offerings. Education providers can use online or face-to-face training, focused on contraindications and screening, determination of BFR pressure, cuff application, and the safety of BFR.

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